oxygen PRESENTS

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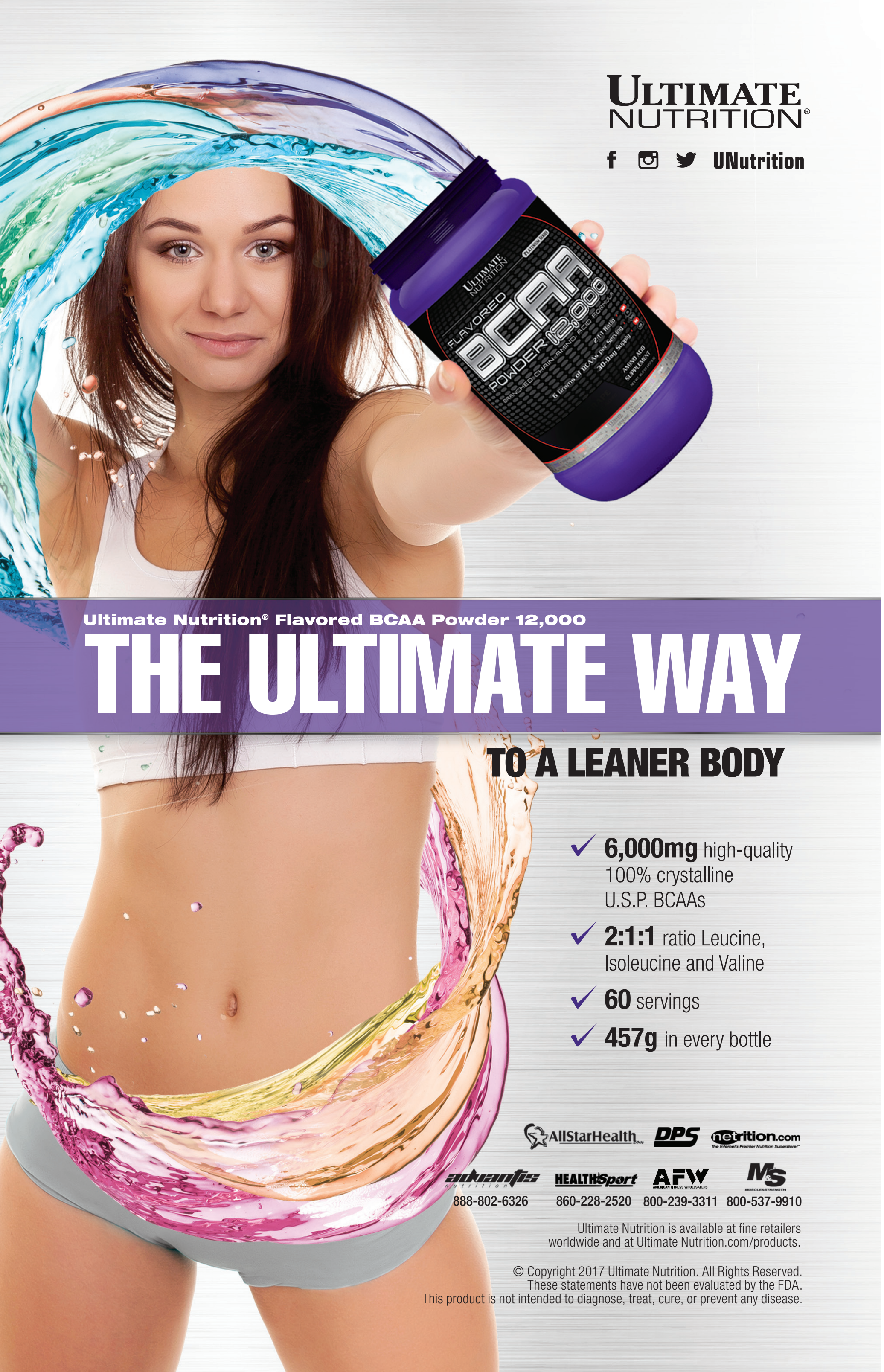
Supplement Solutions



Introduction

You've heard the claims: If you eat a well-rounded diet, you don't need supplements. But even if you're a health-focused, trained woman who (almost) always eats clean, it can be difficult to maintain a proper nutritional profile. One reason is the diminishing quality of our food supply. Modern conventional farming practices deplete the nutrient density of soil and reduce the micronutrient content of animal-based products such as beef, milk, chicken and eggs. If you eat organic, that will certainly net you more vitamins and minerals, but because you work out, you may still be at risk. Research shows that athletes who train intensely routinely lose many critical vitamins and minerals, making supplements an absolute necessity.

You train hard and eat right, but sometimes you need a little extra boost to help you get the results you desire. And a deficiency could keep you out of the gym. Incorporating some of the nutrients listed in this booklet in your daily routine could improve your body's chances at gaining lean muscle, losing body fat or just staying healthy. Whatever your goals, here are some of the must-have supplements to aid your quest.

— VICKI BAKER, SPECIAL PROJECTS EDITOR

Antioxidants help prevent or delay cell damage.

Ante-up with antioxidants

Antioxidants are compounds that can prevent or slow the oxidative damage that contributes to chronic disease. They quench free radicals, which damage cells, including muscle cells. The four listed here also perform other critical functions in the body, including aiding training and fat loss. Add them to your supplement regimen for a leaner, stronger and healthier you.

• BETA CAROTENE

Beta carotene is a form of vitamin A, which plays an important role in eyesight and is involved in maintaining healthy skin, teeth and bones. It can help fight off infection and assist in the growth and repair of body tissues, including muscle.

Research shows that vitamin A is critical for energy production, so if you find yourself fatigued during your daily routine, consider your body's vitamin A status. You also should be sure your vitamin A levels are up to par if you want to maximize energy levels during workouts.

Because vitamin A can be stored in fat cells, its levels in the body can build up to toxic amounts. Supplement with beta carotene, a nontoxic source of vitamin A.

• SELENIUM

This trace mineral is a powerful antioxidant that combats the damage free radicals can cause the body. It's implicated in thyroid health, assisting with the metabolism of iodine and the production of critical thyroid hormones. Because the thyroid controls the body's metabolism, keeping it in good working order is essential to maintaining a lean physique. Selenium also helps limit damage to muscle and other tissues and helps speed recovery.

• VITAMIN C

This powerful antioxidant is involved in the synthesis of hormones, amino acids and collagen. It also protects immune-system cells from damage, allowing them to work more efficiently.

Vitamin C strengthens capillaries and other blood vessels and plays a role in healing injuries. And of course, as an antioxidant, it scavenges free radicals. Studies have shown that vitamin C also can enhance nitric-oxide production (especially when taken in conjunction with L-arginine) and even fat oxidation, which can lead to greater fat loss. Finally, this antioxidant also has strong cortisol-lowering effects. Jolts of cortisol from stress or workouts spur the catabolism of proteins, resulting in poor recovery and loss of muscle tissue.

Be sure to take vitamin C with food. Because the body cannot store the vitamin, it's better to take it at least twice a day to keep levels up.

• VITAMIN E

Vitamin E is a fat-soluble vitamin that also happens to be a powerful antioxidant. Vitamin E is absolutely critical for muscle recovery. Researchers discovered that tears in muscle membranes would not heal unless the muscle cells were treated with vitamin E. Free radicals can prevent muscle tears from healing, and vitamin E's free-radical-fighting antioxidant properties prevent further damage to cells, including muscle cells.

Look for the natural form of vitamin E, d-alpha tocopherol, which is over two times more bioactive than the synthetic form, dl-alpha tocopherol.

Power up with protein

It's a fact that protein intake is essential for adding lean muscle. Gaining muscle requires positive protein balance within the body, meaning that it's important to consume more protein than is broken down during exercise. The general rule of thumb for active people is 1 gram of protein per pound of bodyweight per day, taking in no more than 35 grams at a time.

After a meal, the digestive system breaks protein into its separate amino acids. Animal proteins such as chicken, beef and dairy contain all the essential amino acids that the body can't manufacture on its own. Vegetable proteins tend to have lower levels of amino acids than meat and dairy, but don't count them out — it's important to get sufficient amounts of all the amino acids from different sources.

● CASEIN

Casein makes up 80 percent of the protein content of cow's milk. Known as a time-released protein, it's digested more slowly than whey and provides muscles with a steady stream of amino acids for a longer period (up to seven hours) — such as between meals and while you sleep. Micellar casein is the slowest-digesting form of casein.

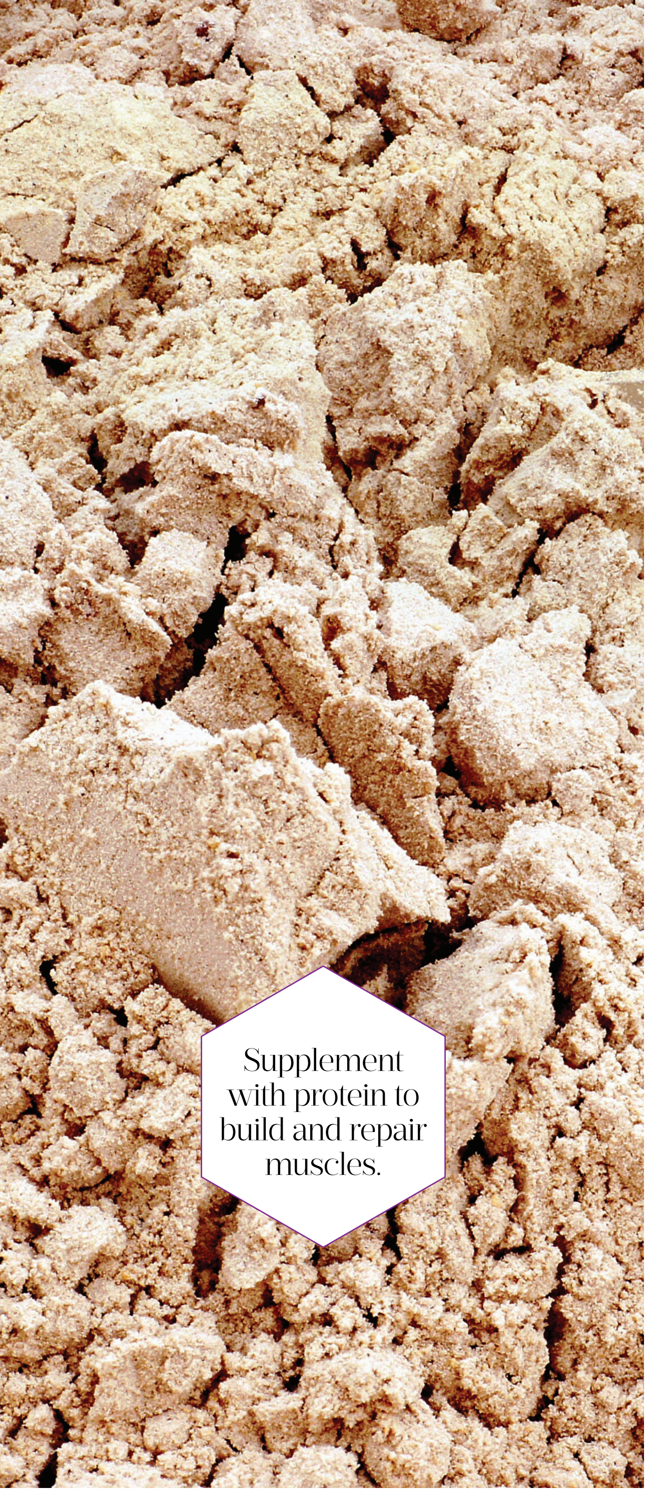
● EGG

Egg-white protein yields a nutrient-rich source of high-quality protein, and it's virtually devoid of carbs and fat. It's absorbed by the body at a rate somewhere between fast-digesting whey protein and slow-digesting casein, so it's safe to crack open a tub at pretty much any time of the day.

● HEMP

Hemp is a heart-healthy plant protein known for helping keep cholesterol levels in check. Hemp's cardio-protective properties can be attributed to its polyunsaturated fatty-acid content, specifically omega-3 fatty acids, which have anti-inflammatory properties. Another bonus to hemp-based proteins is that they're easily digested and contain significant amounts of all the essential amino acids. Preliminary research of hemp-seed protein shows that it may be able to stave off fatigue during exercise, as well as improve immune activity.

Although hemp protein powder supplies only about 50 percent protein, it is rich in branched-chain amino acids, arginine, essential fatty acids and fiber. The fiber content (about 8 grams per serving) and fats slow digestion of the protein, so it's a good choice for between meals and while you sleep.



Supplement with protein to build and repair muscles.

● PEA

Pea protein is a very healthy vegetable protein that can be used alone or combined with protein from dairy sources. Pea protein is naturally imbued with the essential amino acids, including BCAAs, and it has especially high levels of arginine, which has proved its worth as an ergogenic aid, helping delay exercise-induced fatigue and enhancing performance.

Pea protein is especially good for those with allergy concerns. (Dairy and soy are both on the list of common allergens.) It's also vegan and free of gluten, lactose and cholesterol.

● RICE

After a specialized process that treats brown rice with enzymes to separate its carbs from protein, this fibrous little grain boasts roughly 70 percent protein in powder form. It also contains about four times more arginine than whey. Take it before your workout and you'll get a nice boost of pump-inducing NO, as well as glutamine. And it's free of gluten, wheat, egg, milk and soy, making it ideal for vegans and those with food allergies.

● SOY

Soy is probably the most well-known vegetable protein source, and like dairy proteins, it's a complete protein. In the gym, soy has proved itself for increasing lean muscle. One study showed that soy stimulated muscle protein synthesis during exercise and at rest. Another study showed that dieters on a high-soy, low-fat diet lost weight and maintained lean mass over a six-month period.

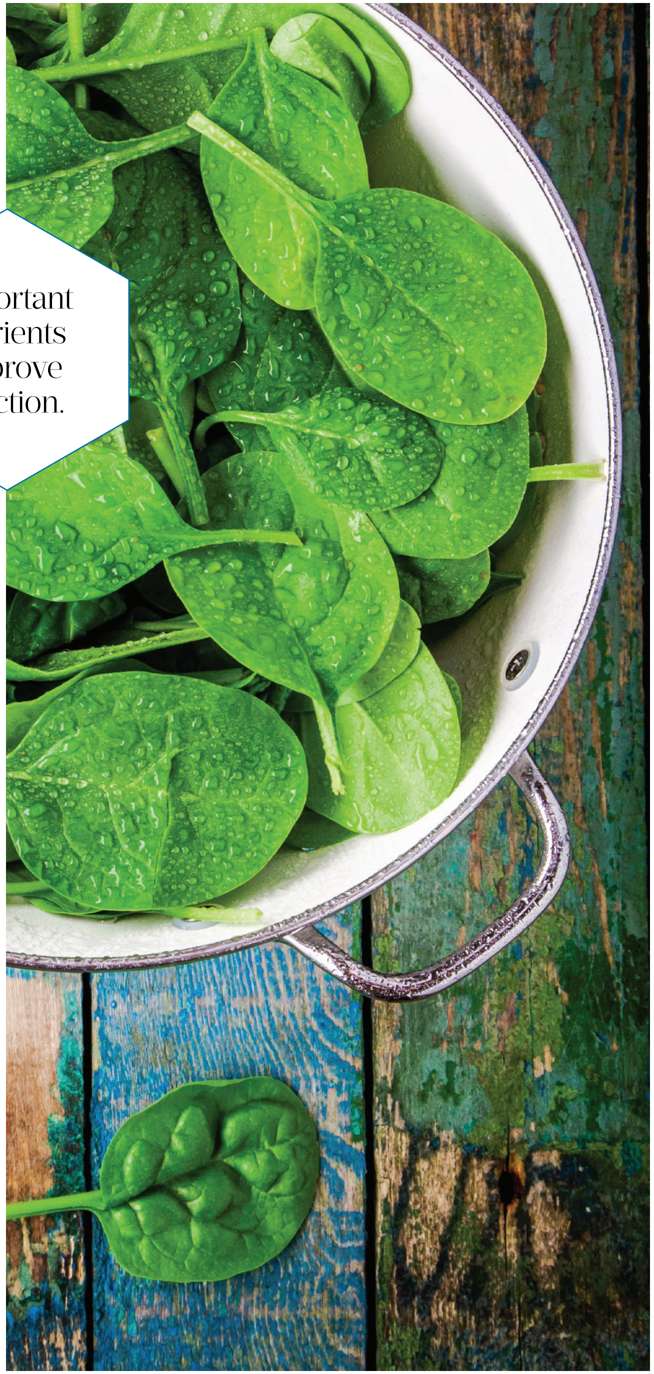
In addition to improving body composition, soy, because it lowers cholesterol, is recommended as a healthy substitute for higher-fat animal proteins. Soy intake seems to have positive effects on bone density, mostly because of its isoflavones. Also known as phytoestrogens, isoflavones actually have mild estrogen-like activity — one reason women with a family or personal history of estrogen-positive breast cancer shy away from soy. As for the threat to women at a higher risk of breast cancer, the research is inconclusive overall, according to a report in a 2010 issue of the *Journal of Epidemiology*.

● WHEY

Touted as the gold standard of proteins, whey makes up 20 percent of the protein content of cow's milk. Whey is a fast-digesting protein, meaning it quickly floods the bloodstream with amino acids, usually within 45 minutes of ingestion. This makes it a good protein to consume for speedy recovery and repair of muscle tissue, before and following workouts. Your best bet is to choose whey protein isolate or hydrolysate, the fastest-digesting varieties available.

Top your tank

Intense gym sessions could be depleting your body of other vital nutrients. Keep your body running in top form by getting in more of these dietary fundamentals.



Important nutrients improve function.

BRANCHED-CHAIN AMINO ACIDS

The BCAAs consist of three essential amino acids: leucine, valine and isoleucine. Your body isn't able to create BCAAs, making it necessary to get them from dietary sources instead.

Together, these three amino acids assist in the repair of muscle tissue almost immediately because they bypass the liver for breakdown and are metabolized right in the muscle. In addition to maintaining your existing muscle tissue by staving off cortisol (a catabolic hormone), BCAAs keep your muscles fueled while you train and keep exercise-induced muscle fatigue at bay. Leucine is responsible for protein synthesis, which in turn promotes muscle growth. BCAAs aid in muscle recovery and help reduce soreness.

BCAAs can be taken as a tablet or powder that you can put in your preworkout and/or post-workout shakes, or you can put flavored BCAAs in your water and drink while you train. Between meals, you can drink flavored BCAAs to keep hunger at bay.

CALCIUM

Calcium is essential to bone integrity, hormone secretion, proper nervous system function, muscle contraction and blood vessel function. Although regular exercise has been found to boost bone mineralization, consistently training hard can lead to a decrease in circulating levels of sex steroids (i.e., estrogen). Low levels of sex steroids can lead to an imbalance in calcium absorption. And when your blood calcium levels dip, the risk for low bone-mineral density and stress fractures increases.

CHROMIUM

While the essential mineral chromium has not been shown to boost performance or aid fat loss, it is often deficient in athletes due to a lack of chromium-rich foods (such as potatoes) in the diet. Because of this, supplementing with chromium is not a bad idea.

Chromium is an insulin cofactor that helps transport BCAAs into the muscle. Without it, you could be at risk for uncontrolled blood sugar and insulin resistance. A chromium deficiency will leave you feeling sluggish, fatigued, weak, moody and anxious.

CREATINE

Creatine supplies muscles with fast energy, allowing you to train longer and harder once your body's natural stores of creatine diminish. It holds water within the muscle not under the skin, keeping your muscle bellies full while increasing their size. Supplementing with creatine helps you train harder longer.

Women do not need to "load" creatine, meaning you don't need to take large doses for several

days to "saturate" the muscles, which can cause bloating and stomach upset. Instead, a small amount either preworkout or postworkout is adequate. If your liver and kidneys are healthy, creatine use will not damage those organs. And you don't need a fancy expensive creatine — a simple monohydrate will work perfectly.

● *DHEA*

DHEA, which stands for dehydroepiandrosterone, is a hormone produced by your adrenal glands, which sit on top of your kidneys. DHEA functions as a precursor to male and female sex hormones, including testosterone and estrogen. Precursors are substances that are converted by the body into a hormone.

DHEA is the most prevalent hormone in the body, and production is highest in your 20s — the average adult produces about 25 milligrams of DHEA per day — but that decreases with age. Your adrenals also produce cortisol and adrenaline, so during times of high stress, the body simply can't manufacture enough DHEA to maintain a healthy hormonal balance, causing everything from extreme fatigue to a decrease in lean muscle.

● *GLUTAMINE*

Glutamine is a nonessential amino acid, which simply means that your body makes it. It is the most abundant amino acid within the body. But it is easily depleted by stress, illness or intense workouts and then it becomes a conditionally essential amino acid, meaning that you need to take in foods or supplements to help replenish levels. Glutamine aids in muscle recovery and has been shown to increase levels of growth hormone.

You can add glutamine powder to preworkout and postworkout shakes. Be careful not to heat it (i.e., putting it in your oatmeal before you add hot water) because it will destroy any of the benefits.

● *MAGNESIUM*

Involved in more than 300 biochemical reactions, including muscle contraction, nerve function, hormonal interactions, immune function and bone health, magnesium is also vital for metabolizing carbohydrates, fats and ATP, the energy system used during training. It appears that it helps improve insulin sensitivity in the body by allowing for better blood glucose control, helping you maintain more steady energy levels and perhaps even helps keep your midriff bikini-ready. A shortfall in this mighty mineral can limit your body's ability to generate energy, leading to fatigue, reduced strength and muscle cramps.

Because magnesium is lost through sweat, hard workouts can leech your body of this mighty mineral. In the supplement form, look for magnesium citrate, which is more readily absorbed.

● *OMEGA-3S EPA AND DHA*

Most people associate omega-3s with cardiovascular health, but their benefits go far beyond the heart. The two main omega-3s — eicosapentaenoic acid and docosahexaenoic acid — promote the body's natural anti-inflammatory response, a normal function that is behind the relief of many chronic conditions but that is often inhibited by poor nutrition. Adequate consumption of these essential fatty acids not only benefits a healthy heart and brain but also supports joint mobility and a normally functioning immune response.

POTASSIUM

Potassium is an essential mineral and electrolyte that promotes proper cell, nerve, brain, heart, kidney and muscle function. When potassium levels take a nosedive, the result can be fatigue, decreased muscle strength, muscle cramps, mood changes, irregular heartbeat and gastrointestinal disturbances, such as bloating and constipation.

Get adequate intake, but avoid over-supplementation, which can interfere with kidney function. Read labels and maintain proper hydration. Food sources include bananas, avocados, almonds and sweet potatoes.

VITAMIN D

This vitamin plays a crucial role in bone mineralization through the regulation of calcium and phosphorus. It promotes the absorption of calcium, integrates calcium into bone tissue, and helps maintain bone density and strength. It also regulates immune and neuromuscular function. A deficiency in D can result in bone loss, muscle weakness and compromised immune function.

Past studies have drawn a connection between vitamin D levels and improved athletic performance, reduced belly fat and improved muscle building. In women, low levels of vitamin D have been linked to an accumulation of fat in muscles, affecting not only appearance but also strength.

Look for labels that say D3, the recommended and natural form that the body makes from sunlight (it metabolizes better in the body).

ZINC

This mineral is critical in the growth, building and repair of muscle tissue, as well as energy production and immune status. It's also essential for thyroid hormone and insulin function. Inadequate intake of zinc can lead to impaired immune function. Your metabolic rate also slows, making it harder to burn off unwanted body fat.

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